Quick DFS

On the infinite square grid a *cell* is a pair of integers (x, y); two cells are *adjacent* if they differ by 1 in exactly one coordinate.

A **region** R is given as the union of n axis-parallel rectangles; the i-th of them consists of all cells (x, y) with $x_{i,1} \le x \le x_{i,2}$ and $y_{i,1} \le y \le y_{i,2}$. (Note that the rectangles are not necessarily disjoint.)

Every region in this task has the following three properties:

- R contains the origin: $(0, 0) \in R$;
- R is **connected**: any two of its cells are joined by a path of adjacent cells that belong to R;
- R is **column-convex**: whenever $(x, y_1) \in R$ and $(x, y_2) \in R$, also $(x, y) \in R$ for every y between y_1 and y_2. In other words, no column of R has a hole.

A robot explores the region using depth-first search, started by `visit(0, 0)`:

```
visit(x, y):
    append (x, y) to the visit order and mark it visited
    # try going up, down, right, and then left, in this specific order:
    for (dx, dy) in [ (0, +1), (0, -1), (+1, 0), (-1, 0) ]:
        if (x + dx, y + dy) belongs to R and is not visited yet:
            visit(x + dx, y + dy)
```

Since R is connected, every cell is appended exactly once, so the visit order is a permutation of the cells of R. The **time** of a cell is its position in that order, indexing from 1.

Task

Answer q queries of two kinds:

- `1 k`: output the coordinates of the cell visited at time k
- `2 x y`: output the time at which the cell (x, y) is visited.

Input format

The first line of the input contains the number t of test cases. The specified number of test cases follows, one after another. Each test case looks as follows:

- The first line contains the integers n and q.
- Each of the next n lines contains four integers $x_{i,1}$, $y_{i,1}$, $x_{i,2}$, $y_{i,2}$ describing one rectangle ($x_{i,1} \le x_{i,2}$, $y_{i,1} \le y_{i,2}$).
- Each of the next q lines contains one query, in one of the two forms above.

All coordinates are between -10^9 and 10^9.

The rectangles are always chosen so that their union R has the three properties listed above.

In a query of the first kind $1 \le k \le |R|$, where $|R|$ is the number of cells of the region.

In a query of the second kind it is guaranteed that $(x, y) \in R$.

Output format

For every query, in the order in which they are given, output one line: two integers (the coordinates of the cell) for a query of the first kind, one integer (the time) for a query of the second kind. Both answers are uniquely determined.

Subproblem Q1 (12 points, public)

Input file: Q1.in (/static/finals2026/Q1.fc3076c5fb24e7155ad927125cc4f3a5.in)

Constraints: $t = 100$; in each test case $n \le 200$, $q \le 200$ and the region has at most $2 \cdot 10^5$ cells.

Subproblem Q2 (18 points, secret)

Input file: Q2.in (/static/finals2026/Q2.1b50dcbc51e0ef48f0d41ce5200c8bee.in)

Constraints: $t = 100$; in each test case $n = 1$ (the region is a single rectangle) and $q \le 2000$.

Subproblem Q3 (30 points, secret)

Input file: Q3.in (/static/finals2026/Q3.88d0114a2494d986c70e67849d93a0d9.in)

Constraints: $t = 100$; in each test case $n \le 2000$ and $q \le 2000$; the sum of n over all test cases is at most 10^5 and the sum of q is at most $1.2 \cdot 10^5$.

Subproblem Q4 (40 points, secret)

Input file: Q4.in (/static/finals2026/Q4.baaca6ae8fa1745df6d3e73303e46b53.in)

Constraints: $t = 40$; in each test case $n \le 10^5$ and $q \le 10^5$; the sum of n over all test cases is at most $3 \cdot 10^5$ and the sum of q is at most $2 \cdot 10^5$.

Examples

input

```
2
4 6
-1 -1 1 2
2 2 3 5
4 0 5 3
-1 2 5 2
1 1
1 9
1 13
1 28
2 -1 2
2 4 1
1 3
0 0 3 1
1 8
2 3 0
2 0 1
```

output

```
0 0
-1 -1
2 2
4 1
12
28
3 0
8
2
```

The region of the first test case, with the time of visiting given in every cell of R:

```
y\x|  -1   0   1   2   3   4   5
---+----------------------------
 5 |   .   .   .  16  17   .   .
 4 |   .   .   .  15  18   .   .
 3 |   .   .   .  14  19  22  23
 2 |  12   3   4  13  20  21  24
 1 |  11   2   5   .   .  28  25
 0 |  10   1   6   .   .  27  26
-1 |   9   8   7   .   .   .   .
```

Note that the region R for this test case would remain exactly the same if we only used the first three of the four rectangles given in the input.